

# Jiwon Heo

 [Homepage](#)

 [4865sj@gmail.com](mailto:4865sj@gmail.com)

 [linkedin.com/in/4865sj/](https://www.linkedin.com/in/4865sj/)

 [github.com/4865sj](https://github.com/4865sj)

## Education

### Korea University

March 2019 - August 2024

*Bachelor*

*Seoul, Republic of Korea*

- **Physics:** 1st Major, GPA 4.03/4.50
- **Mathematics:** Double Major, GPA 4.21/4.50
- **Economics:** Minor, GPA 4.08/4.50
- Total GPA 4.07/4.50

### KAIST

September 2024 - Now

*Ph.D.*

*Daejeon, Republic of Korea*

## Experience

### Signal Processing & Advanced Intelligence (SPAI) LAB (Prof. Jiwon Yoon)

July 2022 – December 2023

*Intern*

*Seoul, Republic of Korea*

### Quantum information theory (QIT) LAB (Prof. Changhun Oh)

September 2024 - Now

*Ph.D.*

*Daejeon, Republic of Korea*

## Paper list

### Classical simulation of a quantum circuit with noisy magic inputs

- Proposed a novel classical simulation algorithm for simulating a quantum circuit with noisy magic inputs.
- Our algorithm is established in a general setting; hence, we apply our algorithm to several scenarios and analyze them.

### Accelerating secure permutation: Application to matrix algebra

- Proposed a new, rapid method for multiplying ciphertext by a specialized form of plaintext, leveraging the algebraic properties of ciphertext rather than traditional homomorphic multiplication
- Exhibited theoretical and experimental improvements and availability by applying it to matrix multiplication.

### Speed-up of Data Analysis with Kernel Trick in Encrypted Domain

### Fast private location-based information retrieval over the torus

## Patent list

### 캐노니컬 임베딩을 사용하는 원분다항식몫체 기반의 암호문에서 특정 원소를 빠르게 추출하는 장치 및 방법

*Domestic patent (Application no. 10-2023-0124483) (Proceeding)*

- Proposed a new efficient operation about extracting specific elements from ciphertext based on cyclotomic polynomial quotient field, utilizing its algebraic properties.
- Led proactively patent application process, including drafting transmittal letter and meeting with patent attorney.

## Core projects

### Numerical analysis for ODE | Python

PHYS233

- Conducted numerical analysis of ordinary differential equations corresponding to classical dynamics with friction.

### Hypothesis testing based on ML and its applications to physics experiments | Python, Tensorflow

PHYS466

- Designed and implemented hypothesis testing based on machine learning and apply to real experimental data.

### Implementation of hypothesis testing | Python

PHYS303

- Embodied t-test and applied it to real data in order to determine whether experimental data can explain its corresponding theory.

### Implementation of Bayesian statistics | Python

STAT404

- Implemented basic concepts of Bayesian statistics and applied them to real data.